THE $L/2L$ PARALLEL AND HOW OUR WORK BUILDS ON IT A NOTE ACCOMPANYING *VERIFICATION AND COMMENTARY: THE BAEZ–EGAN EMBEDDING OF THE LEECH LATTICE IN THE EXCEPTIONAL JORDAN ALGEBRA*

CLAUDE OPUS 4.7

Abstract. This note accompanies the Verification and Commentary of 16 April 2026 on the Baez–Egan embedding (`paper/Baez-Egan_Leech_Verification.pdf`, 8 pp). It sets out an explicit lattice-algebraic parallel between Baez/Egan's "key lemma" (Part 9, $L_1 \cap L_2 = 2L_0$) and Wilson's identity ($L\bar{s} \cap Ls = 2L$), and how Appendix A of our paper *An order on the Leech lattice from a $\mathbb{Z}_3$-symmetric triple-octonion product* (`paper/main.tex`, v8), "The closure criterion modulo 2", builds on that shared identity. It credits the shared lattice-algebraic foundation and the prior work on $L/2L$, and states what our paper adds.

1. The shared key lemma

Both Wilson and Baez/Egan rest on the same lattice-algebraic identity, in different notations:

Baez/Egan (Part 9; §4 of the Verification and Commentary):	$L_1 \cap L_2 = 2L_0$,
Wilson (2009, Section 2):	$L\bar{s} \cap Ls = 2L$.

In Baez/Egan's setup, $L_0 \cong E_8$ is self-dual; $L_1, L_2 \cong \sqrt{2}\,E_8$ are sublattices of L_0 arising from a common rotation R, and the duality argument $\mathrm{dual}(L_1 \cap L_2) = \frac{1}{2}L_1 + \frac{1}{2}L_2 = \frac{1}{2}L_0$ gives $L_1 \cap L_2 = 2L_0$ (Egan).

In Wilson's setup, $L = D_8^+$ is the octonion-realized E_8; the sublattices $L\bar{s}$ and Ls are the images of L under right multiplication by $\bar{s}$ and s respectively (so each is isomorphic to $\sqrt{2}\,E_8$, since $N(s) = N(\bar{s}) = 2$), and Wilson states the identity directly.

The two identities are the same algebraic fact about E_8 and a pair of $\sqrt{2}\,E_8$ sublattices: Wilson's version is the octonion-specific instance; Baez/Egan's is the geometric stripping of it (Egan's Part 9). Either form was already in the literature before this project began.

2. What Baez/Egan use it for, and what they did not do

Baez/Egan use the key lemma in Part 9 as a step in establishing that the lattice

$$L_L \;=\; \{(a,b,c) \in L_0^3 : a,b,c \in L_0;\; a{+}b, a{+}c, b{+}c \in L_1;\; a{+}b{+}c \in L_2\}$$

is isometric to $\sqrt{2}\,\Lambda$. The identity $L_1 \cap L_2 = 2L_0$ is used in showing that $L_L/\sqrt{2}$ is (i) self-dual and (ii) without vectors of squared norm 2. This is the use in Egan's Part 9 of the original n-Category Café series.

Baez/Egan do *not*, anywhere in the Part 9 / Part 10 sequence, descend the octonion product on $L_0 = E_8$ modulo $2L_0$ to read $L_0/2L_0$ as an $\mathbb{F}_2$-bilinear algebra and analyze the sublattice images $L_1/2L_0$, $L_2/2L_0$ in that algebra. The key lemma is used at the level of lattices in $\mathbb{O}^3$, not at the level of an $\mathbb{F}_2$-algebra structure inside $L_0/2L_0$.

Date: 24 May 2026; updated 24 October 2026.

3. What our Appendix A builds on the same identity

Appendix A of our paper ("The closure criterion modulo 2", `paper/main.tex`, v8) uses the same identity, in Wilson's form,

$$L\bar{s} \cap Ls \;=\; 2L, \tag{†}$$

as the foundation for an explicitly algebraic reading. The steps:

(1) From (†), $2L \subseteq L\bar{s}$ and $2L \subseteq Ls$; applying σ (which fixes $2L$, since $\sigma(L) = L$ by Lemma 4.1 in our paper), $2L \subseteq \sigma(L\bar{s})$ and $2L \subseteq \sigma(Ls)$.
(2) $L \cdot L \subseteq L$ (Coxeter; classical) and $\mathbb{Z}$-bilinearity descend the octonion product to an $\mathbb{F}_2$-bilinear product $\bar{\mu}$ on $\bar{L} := L/2L \cong \mathbb{F}_2^8$.
(3) The sublattices $\sigma(L\bar{s})$ and $\sigma(Ls)$ are full preimages of 4-dimensional $\mathbb{F}_2$-subspaces $V := \sigma(L\bar{s})/2L$ and $W := \sigma(Ls)/2L$, and from $\sigma(L\bar{s}) + \sigma(Ls) = L$ together with (†) (applied after σ), $\bar{L} = V \oplus W$.
(4) The closure statements of Lemmas 4.3 and 4.4 in our paper become, mod $2L$: V is a left ideal of $\bar{L}$, two-sided since $\sigma(L\bar{s})$ is a two-sided ideal of L (Remark 4.5); W is a subalgebra of $\bar{L}$.
(5) Under the natural plus-type quadratic form $q(v + 2L) := N(v)/2 \pmod 2$ on $\bar{L}$, direct verification shows that both V and W are totally isotropic, hence Lagrangians of $(\bar{L}, q)$; the decomposition $\bar{L} = V \oplus W$ is a Witt decomposition into a complementary pair of Lagrangians.

Step (1) uses identity (†) directly. Step (5) reads $\bar{L} = V \oplus W$ as a polarization of $(\bar{L}, q)$, an ordered pair of complementary Lagrangians. For Wilson's pair $(L\bar{s}/2L, Ls/2L)$ that reading is prior work (Section 4), and σ, an isometry of L, maps that pair to (V, W). What our paper adds is the product $\bar{\mu}$ of step (2), under which V and W are a two-sided ideal and a subalgebra (step (4)). Baez/Egan in Parts 9–10 do not take this $\mathbb{F}_2$-algebra step.

4. Attribution and overlap

What is shared with Baez/Egan:

- The lattice-algebraic identity $L_1 \cap L_2 = 2L_0$ (Egan's Part 9 key lemma), equivalent to Wilson's $L\bar{s} \cap Ls = 2L$. Egan's duality-based proof of this is in §4 of the Verification and Commentary of 16 April 2026; Section 1 above summarizes it in one line.
- The role of this identity in setting up a Leech-lattice construction inside L_0^3 via three sum-of-components conditions (Wilson 2009 in octonion language; Egan Part 9 in lattice language).

The polarization reading of $\bar{L}$ is prior work. Lepowsky and Meurman (1982, §2) identify the lattices between $2L$ and L that are isometric to $\sqrt{2}\,L$ with the preimages of the Lagrangians of $(\bar{L}, q)$, and construct a lattice isometric to $\sqrt{2}\,\Lambda$ from any polarization; their introduction notes that work of Tits contains what amounts to the same construction. Egan's construction is of this form. In Baez's series, Tim Silverman arrived at the identification of these sublattices with the Lagrangians independently (Part 12). Silverman's comments of 16 and 19 February 2016 on Part 12 read Egan's construction modulo 2 and find that Wilson's pair $(L\bar{s}/2L, Ls/2L)$ is a polarization. David Calderbank related Egan's construction to that of Lepowsky and Meurman (comment of 2 April 2023 on Part 9).

The main paper (`paper/main.tex`, v8) states the sublattice identities in Wilson's setting (Section 2.3) and Egan's construction in Section 6, for sublattices $L_1, L_2 \subset L$ isometric to $\sqrt{2}\,L$ with $L_1 \cap L_2 = 2L$; Wilson's lattice is the case $(L_1, L_2) = (L\bar{s}, Ls)$. The paragraph "Relation to prior work" of its Appendix A gives the credits above. This note sets out the parallel between Egan's key lemma and Wilson's identity (†), on which Appendix A builds.

5. What it would take to close the gap further

A natural follow-up question: does Baez/Egan's framing extend to the $\mathbb{F}_2$-algebra picture of Appendix A? In their setup the generalized lattice $L_L \subset L_0^3$ is constructed without explicit octonion multiplications; once those multiplications are brought back in (as in Wilson 2009, and as required for our $\bar{\mu}$ descent), the $\mathbb{F}_2$-algebra picture is available. Egan's Part 9 chooses to abstract away from octonion multiplication; that abstraction is what makes our $\mathbb{F}_2$-algebra reading absent from his account. In this sense, the present work is a continuation of the Wilson-side of the Wilson–Egan parallel, not the Egan-side.

Conversely, the Part 10 construction (lifting Λ to the 27-dimensional Jordan-algebra ambient $\widetilde{\Lambda} \subset \mathfrak{h}_3(\mathbb{O})$, closed under $2(XY + YX)$) is a different direction: it embeds Λ into a 27-dimensional ring, whereas our mod-2 picture works within the 8-dimensional $L/2L$ inherited from L alone. The two routes are not in competition; they are different uses of the same shared key lemma.

References

- Verification and Commentary: `paper/Baez-Egan_Leech_Verification.pdf`, 16 April 2026 (Claude, Anthropic).
- Our main paper: *An order on the Leech lattice from a $\mathbb{Z}_3$-symmetric triple-octonion product*, v8 (`paper/main.tex`), Appendix A "The closure criterion modulo 2" and Section 6 "Related work".
- Wilson, R. A., *Octonions and the Leech lattice*, J. Algebra **322** (2009), 2186–2190.
- Baez, J. C. (with G. Egan, T. Silverman and others), *Integral octonions* (parts 9, 10 and 12), n-Category Café, November–December 2014 (parts 9 and 10) and 31 January 2016 (part 12, a guest post by T. Silverman); comments by T. Silverman of 16 and 19 February 2016 on part 12, and by D. Calderbank of 2 April 2023 on part 9.
- Lepowsky, J., and Meurman, A., *An E_8-approach to the Leech lattice and the Conway group*, J. Algebra **77** (1982), 484–504.

Anthropic, at the direction of Jens Köplinger